In the Name of God

Deep Generative Models (25732-1)

Homework 2

Fall 2026

Department of Electrical Engineering

Sharif University of Technology

Instructor: Dr. Sajjad Amini

Deadline: Wednesday, Aban 27th, 23:59

Problem 1 . Conditional Variational Autoencoders

Consider a Conditional Variational Autoencoder (CVAE) with an observed variable $\boldsymbol{x}$, a conditioning variable $\boldsymbol{u}$, and a latent variable $\boldsymbol{z}$. For example, $\boldsymbol{u}$ may represent a digit label and $\boldsymbol{x}$ a corresponding handwritten digit image.

The model is defined as

$$p_\theta(\boldsymbol{x}, \boldsymbol{z} \mid \boldsymbol{u}) = p_\theta(\boldsymbol{x} \mid \boldsymbol{z}, \boldsymbol{u}) p_\theta(\boldsymbol{z} \mid \boldsymbol{u})$$

Our goal is to learn the conditional distribution of observations given $\boldsymbol{u}$ by maximizing $\log p_\theta(\boldsymbol{x} \mid \boldsymbol{u})$. However, the exact posterior $p_\theta(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u})$ is generally intractable. We therefore use a variational distribution $q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u})$ to approximate it.

1.1. Deriving the Conditional ELBO

Using Jensen's inequality, prove that :

$$\log p_\theta(\boldsymbol{x} \mid \boldsymbol{u}) \geq \underbrace{\mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x},\boldsymbol{u})}\left[\log \frac{p_\theta(\boldsymbol{x}, \boldsymbol{z} \mid \boldsymbol{u})}{q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u})}\right]}_{\mathcal{L}(\theta,\phi;\boldsymbol{x},\boldsymbol{u})}.$$

Now give a second, independent proof of the same lower bound.

Starting from the definition of KL divergence between $q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u})$ and $p_\theta(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u})$, prove that:

$$\log p_\theta(\boldsymbol{x} \mid \boldsymbol{u}) = \mathcal{L}(\theta, \phi; \boldsymbol{x}, \boldsymbol{u}) + \mathrm{KL}(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u}) \,\|\, p_\theta(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u})),$$

where

$$\mathcal{L}(\theta, \phi; \boldsymbol{x}, \boldsymbol{u}) = \mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x},\boldsymbol{u})}\left[\log \frac{p_\theta(\boldsymbol{x}, \boldsymbol{z} \mid \boldsymbol{u})}{q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u})}\right].$$

Then, show that

$$\mathcal{L}(\theta, \phi; \boldsymbol{x}, \boldsymbol{u}) \leq \log p_\theta(\boldsymbol{x} \mid \boldsymbol{u}),$$

and prove that equality holds if and only if $q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u}) = p_\theta(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u})$ almost everywhere.

1.2 . Decomposing the ELBO

Prove that the ELBO can be decomposed as :

$$\mathcal{L}(\theta, \phi; \boldsymbol{x}, \boldsymbol{u}) = \underbrace{\mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x},\boldsymbol{u})} \left[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z}, \boldsymbol{u})\right]}_{\text{Reconstruction}} - \underbrace{\mathrm{KL}(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u}) \,\|\, p_\theta(\boldsymbol{z} \mid \boldsymbol{u}))}_{\text{Regularization}}.$$

Briefly explain the role of each term.

Suppose the variational posterior matches the conditional prior:

$$q_\phi(z \mid x, u) = p_\theta(z \mid u)$$

Prove that the regularization term vanishes, while the ELBO gap becomes

$$\begin{aligned}\log p_\theta(x \mid u) - \mathcal{L} = \; & \log \mathbb{E}_{p_\theta(z|u)}[p_\theta(x \mid z, u)] \\ & - \mathbb{E}_{p_\theta(z|u)}[\log p_\theta(x \mid z, u)].\end{aligned}$$

Show that this gap is strictly positive whenever $p_\theta(x \mid z, u)$ is not constant with respect to z.

Now suppose the decoder ignores the latent variable, so that

$$p_\theta(\boldsymbol{x} \mid \boldsymbol{z}, \boldsymbol{u}) = p_\theta(\boldsymbol{x} \mid \boldsymbol{u}).$$

Prove that

$$p_\theta(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u}) = p_\theta(\boldsymbol{z} \mid \boldsymbol{u}).$$

Show that if the variational posterior also matches the conditional prior, both the regularization term and the ELBO gap vanish. Explain how this corresponds to posterior collapse.

1.3 . Closed-Form KL Divergence

Assume that the approximate posterior and conditional prior are Gaussian distributions with diagonal covariance matrices:

$$\begin{aligned} q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u}) &= \mathcal{N}(\boldsymbol{\mu}_q, \mathrm{diag}(\boldsymbol{\sigma}_q^2)), \\ p_\theta(\boldsymbol{z} \mid \boldsymbol{u}) &= \mathcal{N}(\boldsymbol{\mu}_p, \mathrm{diag}(\boldsymbol{\sigma}_p^2)), \end{aligned}$$

where $\boldsymbol{z} \in \mathbb{R}^d$.

Derive a closed-form expression for $\mathrm{KL}(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}, \boldsymbol{u}) \,\|\, p_\theta(\boldsymbol{z} \mid \boldsymbol{u}))$ for the general diagonal Gaussian prior and for the standard Gaussian prior $p_\theta(\boldsymbol{z} \mid \boldsymbol{u}) = \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$.

1.4 . Reparameterization Trick

While the KL term in 1.3 admits a closed-form expression, the reconstruction term generally does not. Construct a Monte Carlo estimator of the reconstruction gradient with respect to ϕ using the reparameterization trick. Give the deterministic transformation of $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, justify that it produces the desired Gaussian distribution, and state the noise based reconstruction objective and its chain rule gradient.

Problem 2 . Hierarchical VAEs

In a standard VAE, a single latent variable is used to model the observed data. Hierarchical VAEs extend this framework by introducing multiple dependent latent variables.

Consider a two-layer hierarchical VAE with latent variables $\boldsymbol{z}_1$ and $\boldsymbol{z}_2$. We assume that $\boldsymbol{x}$ is conditionally independent of $\boldsymbol{z}_2$ given $\boldsymbol{z}_1$, i.e., $\boldsymbol{x} \perp\!\!\!\perp \boldsymbol{z}_2 \mid \boldsymbol{z}_1$.

The model is defined as

$$p_\theta(\boldsymbol{x}, \boldsymbol{z}_1, \boldsymbol{z}_2) = p_\theta(\boldsymbol{z}_2)p_\theta(\boldsymbol{z}_1 \mid \boldsymbol{z}_2)p_\theta(\boldsymbol{x} \mid \boldsymbol{z}_1).$$

To approximate the intractable posterior, we use the variational distribution

$$q_\phi(\boldsymbol{z}_1, \boldsymbol{z}_2 \mid \boldsymbol{x}) = q_\phi(\boldsymbol{z}_2 \mid \boldsymbol{x})q_\phi(\boldsymbol{z}_1 \mid \boldsymbol{z}_2, \boldsymbol{x}).$$

2.1 . Hierarchical ELBO

Starting from the KL divergence between the approximate posterior and the true posterior, prove that

$$\log p_\theta(\boldsymbol{x}) = \mathcal{L}_{\mathrm{H}}(\boldsymbol{x}) + \mathrm{KL}(q_\phi(\boldsymbol{z}_1, \boldsymbol{z}_2 \mid \boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}_1, \boldsymbol{z}_2 \mid \boldsymbol{x}))\,,$$

Derive the hierarchical ELBO

$$\mathcal{L}_{\mathrm{H}}(\boldsymbol{x}) = \mathbb{E}_{q_\phi(\boldsymbol{z}_1, \boldsymbol{z}_2 \mid \boldsymbol{x})}\left[\log \frac{p_\theta(\boldsymbol{x}, \boldsymbol{z}_1, \boldsymbol{z}_2)}{q_\phi(\boldsymbol{z}_1, \boldsymbol{z}_2 \mid \boldsymbol{x})}\right]$$

in the following decomposed form:

$$\begin{aligned}\mathcal{L}_{\mathrm{H}}(\boldsymbol{x}) = \mathbb{E}_{q_\phi(\boldsymbol{z}_1, \boldsymbol{z}_2 \mid \boldsymbol{x})}[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z}_1)] \\ - \mathrm{KL}(q_\phi(\boldsymbol{z}_2 \mid \boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}_2)) \\ - \mathbb{E}_{q_\phi(\boldsymbol{z}_2 \mid \boldsymbol{x})}\left[\mathrm{KL}(q_\phi(\boldsymbol{z}_1 \mid \boldsymbol{z}_2, \boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}_1 \mid \boldsymbol{z}_2))\right]\end{aligned}$$

Express $\log p_\theta(\boldsymbol{x}) - \mathcal{L}_{\mathrm{H}}(\boldsymbol{x})$ as the KL divergence between the approximate and true posteriors, and briefly explain why the final KL term is averaged $q_\phi(\boldsymbol{z}_2 \mid \boldsymbol{x})$.

2.2 . Marginal Priors

Prove that the hierarchical ELBO satisfies

$$\mathcal{L}_{\mathrm{H}}(\boldsymbol{x}) = \mathcal{L}_{\mathrm{M}}(\boldsymbol{x}) - \mathbb{E}_{q_\phi(\boldsymbol{z}_1 \mid \boldsymbol{x})}\left[\mathrm{KL}(q_\phi(\boldsymbol{z}_2 \mid \boldsymbol{z}_1, \boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}_2 \mid \boldsymbol{z}_1))\right],$$

where

$$\mathcal{L}_{\mathrm{M}}(\boldsymbol{x}) = \mathbb{E}_{q_\phi(\boldsymbol{z}_1 \mid \boldsymbol{x})}[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z}_1)] - \mathrm{KL}(q_\phi(\boldsymbol{z}_1 \mid \boldsymbol{x}) \,\|\, p_\theta(\boldsymbol{z}_1))\,.$$

Prove that

$$\mathcal{L}_{\mathrm{H}}(\boldsymbol{x}) \leq \mathcal{L}_{\mathrm{M}}(\boldsymbol{x}).$$

Determine the necessary and sufficient condition for equality. For a fixed marginal variational distribution $q_\phi(\boldsymbol{z}_1 \mid \boldsymbol{x})$, characterize the conditional distribution $q(\boldsymbol{z}_2 \mid \boldsymbol{z}_1, \boldsymbol{x})$ that maximizes the hierarchical ELBO.

The marginal prior $p_\theta(\boldsymbol{z}_1)$ may be intractable to evaluate explicitly. For an arbitrary auxiliary distribution $r(\boldsymbol{z}_2 \mid \boldsymbol{z}_1)$, prove that

$$\begin{aligned}\log p_\theta(\boldsymbol{z}_1) = \mathbb{E}_{r(\boldsymbol{z}_2 \mid \boldsymbol{z}_1)}[\log p_\theta(\boldsymbol{z}_1 \mid \boldsymbol{z}_2)] \\ - \mathrm{KL}(r(\boldsymbol{z}_2 \mid \boldsymbol{z}_1) \,\|\, p_\theta(\boldsymbol{z}_2)) \\ + \mathrm{KL}(r(\boldsymbol{z}_2 \mid \boldsymbol{z}_1) \,\|\, p_\theta(\boldsymbol{z}_2 \mid \boldsymbol{z}_1))\,.\end{aligned}$$

Hence, prove the variational representation

$$\log p_\theta(\boldsymbol{z}_1) = \sup_r \Big\{ \mathbb{E}_{r(\boldsymbol{z}_2|\boldsymbol{z}_1)}[\log p_\theta(\boldsymbol{z}_1 \mid \boldsymbol{z}_2)] - \mathrm{KL}(r(\boldsymbol{z}_2 \mid \boldsymbol{z}_1) \,\|\, p_\theta(\boldsymbol{z}_2)) \Big\}.$$

Determine when the supremum is attained.

Problem 3 . Flow-Based Priors in VAEs

Consider a VAE with a prior defined by an invertible flow:

$$\boldsymbol{u} \sim p_0(\boldsymbol{u}) = \mathcal{N}(\boldsymbol{u}; \mathbf{0}, I_d), \qquad \boldsymbol{z} = f_\psi(\boldsymbol{u}), \qquad \boldsymbol{x} \sim p_\theta(\boldsymbol{x} \mid \boldsymbol{z}).$$

Assume that f_ψ and its inverse are continuously differentiable with nonsingular Jacobians. Let $p_\psi(\boldsymbol{z})$ be the induced prior and $q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ the variational posterior. Assume all expressions below are finite.

3.1 . Flow Prior and ELBO

Using change of variables, derive the ELBO for the flow based prior and show that

$$\mathcal{L}_{\theta,\psi}(\boldsymbol{x}) = \mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{z})] - \mathrm{KL}(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p_\psi(\boldsymbol{z}))\,,$$

where

$$\mathrm{KL}(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p_\psi(\boldsymbol{z})) = \mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x})}\Big[\log q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) - \log p_0(f_\psi^{-1}(\boldsymbol{z})) + \log \Big| \det J_{f_\psi}(f_\psi^{-1}(\boldsymbol{z})) \Big| \Big].$$

3.2 . Equivalent Models

Define a model with the standard Gaussian prior $p_0(\boldsymbol{u})$ and decoder

$$\widetilde{p}_{\theta,\psi}(\boldsymbol{x} \mid \boldsymbol{u}) = p_\theta(\boldsymbol{x} \mid f_\psi(\boldsymbol{u})).$$

Prove that

$$p_{\theta,\psi}(\boldsymbol{x}) = \int p_\theta(\boldsymbol{x} \mid f_\psi(\boldsymbol{u})) p_0(\boldsymbol{u})\, d\boldsymbol{u}.$$

For $\boldsymbol{z} \sim q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$, define $\boldsymbol{u} = f_\psi^{-1}(\boldsymbol{z})$. Derive its density

$$r_{\phi,\psi}(\boldsymbol{u} \mid \boldsymbol{x}) = q_\phi(f_\psi(\boldsymbol{u}) \mid \boldsymbol{x}) |\det J_{f_\psi}(\boldsymbol{u})|.$$

Then prove that

$$\mathrm{KL}(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \,\|\, p_\psi(\boldsymbol{z})) = \mathrm{KL}(r_{\phi,\psi}(\boldsymbol{u} \mid \boldsymbol{x}) \,\|\, p_0(\boldsymbol{u}))\,.$$

Hence, show that the ELBOs of the two representations are equal when their variational distributions are related by this transformation.

Now suppose the variational family is restricted to diagonal Gaussian distributions in each representation. Consider

$$f(u_1, u_2) = (u_1, u_2 + u_1^2), \qquad q(\boldsymbol{z} \mid \boldsymbol{x}) = \mathcal{N}(\boldsymbol{z}; \mathbf{0}, I_2).$$

Find the induced density $r(\boldsymbol{u} \mid \boldsymbol{x})$ and prove that it is not a diagonal Gaussian. Use this example to explain why the equivalence of ELBO optimization can break down under this restriction, even though the two generative models are theoretically equivalent.

Problem 4 . Normalizing Flows

Normalizing flows model continuous probability densities using invertible transformations. However, many datasets, such as digital images, consist of discrete observations.

In this problem, we investigate how normalizing flows can be used to model discrete data through dequantization, following the ideas of Flow++.

4.1 . Affine Coupling Layers

Let $\boldsymbol{h} = (\boldsymbol{h}_A, \boldsymbol{h}_B) \in \mathbb{R}^D$ be partitioned into two vectors of dimensions d_A and d_B, with $d_A + d_B = D$.

Consider the affine coupling transformation $f_\theta : \mathbb{R}^D \to \mathbb{R}^D$, defined by

$$\begin{aligned} \boldsymbol{y}_A &= \boldsymbol{h}_A, \\ \boldsymbol{y}_B &= \boldsymbol{h}_B \odot \exp(s_\theta(\boldsymbol{h}_A)) + t_\theta(\boldsymbol{h}_A), \end{aligned}$$

where $s_\theta, t_\theta : \mathbb{R}^{d_A} \to \mathbb{R}^{d_B}$ are differentiable functions, and $\odot$ denotes elementwise multiplication.

Let p_0 be a base probability density on $\mathbb{R}^D$, let $\boldsymbol{h} \sim p_0$ and $\boldsymbol{y} = f_\theta(\boldsymbol{h})$, and denote the density of $\boldsymbol{y}$ by p_θ. Prove that f_θ is invertible and that

$$\log p_\theta(\boldsymbol{y}) = \log p_0(f_\theta^{-1}(\boldsymbol{y})) - \sum_{j=1}^{d_B} s_{\theta,j}(\boldsymbol{y}_A)$$

Give the inverse. Explain why the determinant remains tractable even when s_θ and t_θ are neural networks.

4.2 . From Continuous Densities to Discrete Probabilities

A normalizing flow defines a continuous density $p_\theta(\boldsymbol{y})$ on $\mathbb{R}^D$. However, an image consists of integer-valued pixel intensities.

For simplicity, let $\boldsymbol{x} \in \mathbb{Z}^D$ denote a discrete observation. Define its corresponding quantization bin as

$$B_{\boldsymbol{x}} = \boldsymbol{x} + [0, 1)^D$$

The bins $\{B_{\boldsymbol{x}} : \boldsymbol{x} \in \mathbb{Z}^D\}$ form a partition of $\mathbb{R}^D$ up to their boundaries.

Let $\boldsymbol{y}$ be distributed according to $p_\theta(\boldsymbol{y})$, and define the quantization operation $\boldsymbol{x} = \lfloor \boldsymbol{y} \rfloor$, applied elementwise. Prove that the probability mass assigned to $\boldsymbol{x}$ is

$$P_\theta(\boldsymbol{x}) = \int_{[0,1)^D} p_\theta(\boldsymbol{x} + \boldsymbol{u})\, d\boldsymbol{u}.$$

Prove that P_θ defines a valid probability mass function on $\mathbb{Z}^D$, i.e.,

$$P_\theta(\boldsymbol{x}) \geq 0, \qquad \sum_{\boldsymbol{x} \in \mathbb{Z}^D} P_\theta(\boldsymbol{x}) = 1.$$

Explain the distinction between the density $p_\theta(\boldsymbol{x})$ and the probability mass $P_\theta(\boldsymbol{x})$.

4.3 . Variational Dequantization

Computing the discrete likelihood $P_\theta(\boldsymbol{x})$ requires integrating the continuous density over the entire quantization bin, which is generally intractable for a neural normalizing flow.

To construct a tractable lower bound, introduce a dequantization distribution $q_\phi(\boldsymbol{u} \mid \boldsymbol{x})$, supported on $[0, 1)^D$. Assume that q_ϕ is strictly positive almost everywhere in this bin and that all relevant expectations are finite. Using importance sampling and Jensen's inequality, prove that

$$\log P_\theta(\boldsymbol{x}) \geq \mathbb{E}_{q_\phi(\boldsymbol{u}|\boldsymbol{x})} \left[\log \frac{p_\theta(\boldsymbol{x} + \boldsymbol{u})}{q_\phi(\boldsymbol{u} \mid \boldsymbol{x})} \right].$$

Denote the right-hand side by $\mathcal{L}_{\mathrm{DQ}}(\theta, \phi; \boldsymbol{x})$

Let the differential entropy of the dequantization distribution be defined as

$$H(q_\phi(\cdot \mid \boldsymbol{x})) = -\mathbb{E}_{q_\phi(\boldsymbol{u}|\boldsymbol{x})}[\log q_\phi(\boldsymbol{u} \mid \boldsymbol{x})]$$

Prove that the dequantization objective can be written as

$$\begin{aligned}\mathcal{L}_{\mathrm{DQ}}(\theta, \phi; \boldsymbol{x}) = \mathbb{E}_{q_\phi(\boldsymbol{u}|\boldsymbol{x})}[\log p_\theta(\boldsymbol{x} + \boldsymbol{u})] \\ + H(q_\phi(\cdot \mid \boldsymbol{x})).\end{aligned}$$

Identify the roles of the likelihood and entropy terms.

4.4 . Uniform Dequantization

We now investigate how the choice of $q_\phi(\boldsymbol{u} \mid \boldsymbol{x})$ affects the tightness of the bound.

Define the normalized density

$$r_\theta(\boldsymbol{u} \mid \boldsymbol{x}) = \frac{p_\theta(\boldsymbol{x} + \boldsymbol{u})}{P_\theta(\boldsymbol{x})}, \qquad \boldsymbol{u} \in [0, 1)^D,$$

assuming $P_\theta(\boldsymbol{x}) > 0$.

Starting from the definition of KL divergence, prove the exact identity

$$\log P_\theta(\boldsymbol{x}) - \mathcal{L}_{\mathrm{DQ}}(\theta, \phi; \boldsymbol{x}) = \mathrm{KL}(q_\phi(\boldsymbol{u} \mid \boldsymbol{x}) \,\|\, r_\theta(\boldsymbol{u} \mid \boldsymbol{x}))$$

Hence, determine the distribution q_ϕ for which the dequantization bound is tight.

Consider the uniform dequantization distribution

$$q_{\mathrm{U}}(\boldsymbol{u} \mid \boldsymbol{x}) = \begin{cases} 1, & \boldsymbol{u} \in [0, 1)^D, \\ 0, & \text{otherwise.} \end{cases}$$

Show that the variational objective reduces to

$$\mathcal{L}_{\mathrm{U}}(\theta; \boldsymbol{x}) = \int_{[0,1)^D} \log p_\theta(\boldsymbol{x} + \boldsymbol{u})\, d\boldsymbol{u}$$

Prove that this bound is tight if and only if $p_\theta(\boldsymbol{x} + \boldsymbol{u})$ is constant almost everywhere within the quantization bin.

Instead of fixing q_ϕ to be uniform, suppose it is parameterized by a conditional normalizing flow:

$$\boldsymbol{\epsilon} \sim p_0(\boldsymbol{\epsilon}), \qquad \boldsymbol{u} = g_\phi(\boldsymbol{\epsilon}; \boldsymbol{x}),$$

where, for each fixed $\boldsymbol{x}$, g_ϕ is a differentiable bijection from $\mathbb{R}^D$ to $(0, 1)^D$, with a differentiable inverse and nonzero Jacobian determinant.

Show that the dequantization objective becomes

$$\begin{aligned}\mathcal{L}_{\mathrm{DQ}}(\theta, \phi; \boldsymbol{x}) = \mathbb{E}_{\boldsymbol{\epsilon} \sim p_0}\Big[\log p_\theta(\boldsymbol{x} + g_\phi(\boldsymbol{\epsilon}; \boldsymbol{x})) \\ + \log \left| \det J_{g_\phi}(\boldsymbol{\epsilon}; \boldsymbol{x}) \right| \\ - \log p_0(\boldsymbol{\epsilon}) \Big]\end{aligned}$$

Explain why a learned flow-based dequantization distribution can yield a tighter bound than uniform dequantization.

Problem 5 . Invertible 1×1 Convolutions in Glow

In Glow, invertible 1×1 convolutions replace fixed channel permutations with learnable linear transformations. This allows information to be mixed across channels while preserving invertibility and a tractable Jacobian determinant.

Let $\boldsymbol{X} \in \mathbb{R}^{C \times H \times W}$ be an input tensor with C channels and spatial dimensions $H \times W$. Consider the transformation

$$\boldsymbol{Y}_{:,i,j} = \boldsymbol{W}\boldsymbol{X}_{:,i,j}, \qquad \boldsymbol{W} \in \mathbb{R}^{C \times C},$$

where the same matrix $\boldsymbol{W}$ is applied independently at every spatial location (i, j).

5.1 . Invertibility and Jacobian Determinant

(a) Prove that the transformation is invertible if and only if $\det(\boldsymbol{W}) \neq 0$, and show that its inverse is given by

$$\boldsymbol{X}_{:,i,j} = \boldsymbol{W}^{-1}\boldsymbol{Y}_{:,i,j}.$$

(b) Derive the Jacobian matrix J of the complete transformation and prove that

$$\log|\det J| = HW \log|\det \boldsymbol{W}|$$

5.2 . LU Parameterization

To make the computation of the log-determinant more efficient, Glow parameterizes the weight matrix as

$$\boldsymbol{W} = \boldsymbol{P}\boldsymbol{L}\left(\boldsymbol{U} + \operatorname{diag}(\boldsymbol{s})\right),$$

where $\boldsymbol{P}$ is a fixed permutation matrix, $\boldsymbol{L}$ is lower triangular with ones on its diagonal, $\boldsymbol{U}$ is strictly upper triangular, and $\boldsymbol{s} \in \mathbb{R}^C$ has nonzero entries.

Using the properties of determinants and triangular matrices, prove that

$$\log|\det \boldsymbol{W}| = \sum_{i=1}^{C} \log|s_i|$$

Hence, show that

$$\log|\det J| = HW \sum_{i=1}^{C} \log|s_i|$$